

EDUGENIE

Google Gemini Powered Learning Assistant

1. Brainstorming and Ideation

Project Idea

- EduGenie is a lightweight AI-powered educational assistant designed to simplify learning through generative AI.
- The system focuses on helping learners ask questions, understand difficult concepts, generate quizzes, summarize educational content, and obtain personalized learning recommendations.

Problem Identification

- Students may find complex educational material difficult to understand quickly.
- Large passages can require significant time to revise and extract key information.
- Learners may need structured guidance when progressing from beginner to advanced topics.
- A simple, low-footprint interface can make AI-assisted learning more accessible.

Proposed Solution

- Build a web-based learning assistant with FastAPI as the backend and HTML/CSS as the frontend.
- Use Google Gemini for question answering, summarization, quiz generation, and learning recommendations.
- Use a lightweight local model for concept explanation as described in the project documentation.

Main Use Cases

- Ask a question such as identifying the largest ocean.
- Generate a quiz to check understanding of a topic such as the Pythagoras Theorem.
- Request a structured learning path for a subject such as SQL, progressing from beginner to advanced.

Expected Outcome

- A simple educational platform that turns static learning into a more interactive process.
- A modular foundation that can be expanded with features such as multilingual support, voice interaction, progress tracking, mobile access, and image/PDF input.

Source basis: EduGenie project documentation provided by the user.